

Def. Let W be a vector field on $\mathbb{R}^3$, and V_p be a tangent vector.

A covariant derivative of W with respect to V_p is the tangent vector

$$\nabla_V W \stackrel{\text{def}}{=} W(p+t \cdot V)'(0)_p$$

Lemma. If $W = \sum_{i=1}^3 w_i \cdot e_i$ is a vector field on $\mathbb{R}^3$, then for any tangent vector V_p ,

$$\nabla_V W = \sum_{i=1}^3 V[w_i] \cdot e_i(p).$$

Proof. We have $W(p+t \cdot V) = \sum_{i=1}^3 \underbrace{w_i(p+tV)}_{\text{scalar valued}} \cdot \underbrace{e_i(p+tV)}_{\text{vector valued}}$,

thus, by the definition of directional derivative,

$$\nabla_V W = \left. \frac{d}{dt} W(p+tV) \right|_{t=0} = \sum \left. \frac{d}{dt} w_i(p+tV) \right|_{t=0} \cdot e_i(p) = \sum V[w_i] \cdot e_i(p).$$